Differential Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given that $y = 3$ when $x = 0$.

Separate: $y\,dy = x\,dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y = 3, x = 0$: $\frac{9}{2} = C$. So $y^2 = x^2 + 9$, giving $y = \sqrt{x^2 + 9}$ (taking the positive root since $y = 3 > 0$).

2 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = 2xy$, given that $y = 5$ when $x = 0$.

Separate: $\frac{dy}{y} = 2x\,dx$. Integrate: $\ln|y| = x^2 + C$. So $y = Ae^{x^2}$ for some constant A. Using $y = 5, x = 0$: $A = 5$. Solution: $y = 5e^{x^2}$.

3 **[Foundational]** • **Target: Modeling with differential equations**

A population P grows at a rate proportional to its current size, modeled by $\frac{dP}{dt} = kP$. If the population doubles every 5 years, find k, to three decimal places.

Solving the DE: $P(t) = P_0e^{kt}$. Doubling: $2P_0 = P_0e^{5k}$, so $e^{5k} = 2$, giving $5k = \ln 2$, $k = \frac{\ln 2}{5} \approx 0.139$.

4 **[Foundational]** • **Target: Modeling with differential equations**

Newton's law of cooling states $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve this differential equation for $T(t)$ in terms of the initial temperature T_0.

Separate: $\frac{dT}{T - T_a} = -k\,dt$. Integrate: $\ln|T - T_a| = -kt + C$. So $T - T_a = Ae^{-kt}$. At $t = 0$, $T = T_0$, so $A = T_0 - T_a$. Solution: $T(t) = T_a + (T_0 - T_a)e^{-kt}$.

5 **[Foundational]** • **Target: First-order linear differential equations**

Solve $\frac{dy}{dx} + 2y = 6$ using an integrating factor, given $y = 1$ when $x = 0$.

Integrating factor: $e^{\int 2\,dx} = e^{2x}$. Multiply through: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$, i.e. $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$. Integrate: $ye^{2x} = 3e^{2x} + C$, so $y = 3 + Ce^{-2x}$. Using $y = 1, x = 0$: $1 = 3 + C$, so $C = -2$. Solution: $y = 3 - 2e^{-2x}$.

6 **[Foundational]** • **Target: Euler's method for numerical solutions**

Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for $\frac{dy}{dx} = x + y$, given $y(0) = 1$, showing each step.

Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$. Step 1 ($x_0 = 0, y_0 = 1$): $y_1 = 1 + 0.5(0 + 1) = 1.5$. Step 2 ($x_1 = 0.5, y_1 = 1.5$): $y_2 = 1.5 + 0.5(0.5 + 1.5) = 1.5 + 1 = 2.5$. Estimate: $y(1) \approx 2.5$.

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7 [Foundational] • Target: Second-order context: simple harmonic motion

The differential equation $\frac{d^2x}{dt^2} = -\omega^2 x$ describes simple harmonic motion. Verify that $x(t) = A\cos(\omega t)$ is a solution.

$\frac{dx}{dt} = -A\omega\sin(\omega t)$. $\frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t) = -\omega^2(A\cos(\omega t)) = -\omega^2 x$, confirming $x(t) = A\cos(\omega t)$ satisfies the differential equation.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Family of solution curves and particular solutions

Find the general solution of $\frac{dy}{dx} = 3x^2$, and explain what the arbitrary constant represents geometrically.

Integrating directly: $y = x^3 + C$. The constant C represents a vertical shift — the general solution is a whole family of curves, all with the same shape (since they share the same derivative $3x^2$ at every x), just shifted up or down by different amounts.

9 [Intermediate] • Target: Family of solution curves and particular solutions

A tank drains such that $\frac{dV}{dt} = -k\sqrt{V}$ for some constant $k > 0$, where V is the volume of water. Solve this differential equation for $V(t)$, given $V(0) = V_0$.

Separate: $\frac{dV}{\sqrt{V}} = -k\,dt$, i.e. $V^{-1/2}\,dV = -k\,dt$. Integrate: $2\sqrt{V} = -kt + C$. Using $V(0) = V_0$: $2\sqrt{V_0} = C$. So $2\sqrt{V} = -kt + 2\sqrt{V_0}$, giving $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$, so $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$.

10 [Intermediate] • Target: Verifying a solution to a second-order differential equation

Verify that $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.

$\frac{dy}{dx} = 2e^{2x}$. $\frac{d^2y}{dx^2} = 4e^{2x}$. Substituting: $4e^{2x} - 5(2e^{2x}) + 6e^{2x} = 4e^{2x} - 10e^{2x} + 6e^{2x} = 0e^{2x} = 0$, confirming $y = e^{2x}$ is a solution.

11 [Intermediate] • Target: Slope fields and qualitative solutions

A differential equation is given by $\frac{dy}{dx} = x - y$. Find the gradient of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, and explain what a slope field represents.

At $(0, 0)$: gradient $= 0 - 0 = 0$. At $(1, 0)$: gradient $= 1 - 0 = 1$. At $(1, 1)$: gradient $= 1 - 1 = 0$. A slope field is a diagram that plots a short line segment of the correct gradient at many points in the plane, without solving the equation explicitly — sketching a solution curve then amounts to drawing a curve that follows the direction of the segments at every point it passes through.

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12 **[Intermediate]** • **Target: A logistic growth model**

A population P grows according to the logistic differential equation $\frac{dP}{dt} = kP(10 - P)$. Using the partial fraction decomposition $\frac{1}{P(10-P)} = \frac{1}{10P} + \frac{1}{10(10-P)}$, solve for P in terms of t, given $P(0) = 1$ and leaving k as an unknown constant.

Separate: $\frac{dP}{P(10-P)} = k\,dt$. Integrate the left side using the given decomposition: $\frac{1}{10}\ln|P| - \frac{1}{10}\ln|10 - P| = kt + C$, i.e. $\frac{1}{10}\ln\left|\frac{P}{10-P}\right| = kt + C$. Using $P(0) = 1$: $\frac{1}{10}\ln\left(\frac{1}{9}\right) = C$. So $\ln\left|\frac{P}{10-P}\right| = 10kt + \ln\left(\frac{1}{9}\right)$, giving $\frac{P}{10-P} = \frac{1}{9}e^{10kt}$. Solving for P: $9P = (10 - P)e^{10kt}$, so $P(9 + e^{10kt}) = 10e^{10kt}$, giving $P = \frac{10e^{10kt}}{9 + e^{10kt}}$.

13 **[Intermediate]** • **Target: Checking the order and degree of a differential equation**

State the order of each of the following differential equations: (a) $\frac{dy}{dx} = 3x^2 - 2y$, (b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$, (c) $\frac{d^3y}{dx^3} = \sin x$.

The order of a differential equation is the order of the highest derivative present. (a) Only a first derivative appears, so this is a first-order differential equation. (b) The highest derivative is $\frac{d^2y}{dx^2}$, so this is second-order. (c) The highest derivative is $\frac{d^3y}{dx^3}$, so this is third-order.

14 **[Intermediate]** • **Target: Using Euler's method with a smaller step size**

Use Euler's method with step size $h = 0.25$ to estimate $y(0.5)$ for $\frac{dy}{dx} = 2x$, given $y(0) = 1$, showing each step, and compare with the exact solution $y = x^2 + 1$.

Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$. Step 1 ($x_0 = 0, y_0 = 1$): $y_1 = 1 + 0.25(2 \cdot 0) = 1 + 0 = 1$. Step 2 ($x_1 = 0.25, y_1 = 1$): $y_2 = 1 + 0.25(2 \cdot 0.25) = 1 + 0.125 = 1.125$. Estimate: $y(0.5) \approx 1.125$. Exact solution: $y(0.5) = (0.5)^2 + 1 = 1.25$. The Euler estimate underestimates the true value, since the method uses the derivative only at the start of each interval rather than accounting for how the derivative changes within it.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Using Euler's method with a smaller step size** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$, and describe the family of curves this represents.

Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + C$, so $y = Ax$ for some constant A. Using $y = 6, x = 2$: $6 = 2A$, so $A = 3$. Solution: $y = 3x$. This represents the family of straight lines through the origin; the particular solution satisfying the initial condition is the specific line with slope 3.

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16 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 4.02 The Poisson Distribution

A cup of coffee at 90°C is placed in a room at ambient temperature 20°C. Using Newton's law of cooling, $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 10 minutes the coffee has cooled to 60°C, find the value of k to three decimal places.

Using $T_a = 20$, $T_0 = 90$, and $T(10) = 60$: $60 = 20 + (90 - 20)e^{-10k}$, so $40 = 70e^{-10k}$, giving $e^{-10k} = \frac{4}{7}$. Taking natural logs: $-10k = \ln\left(\frac{4}{7}\right) \approx -0.5596$, so $k \approx 0.056$.

17 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 5.01 Further Integration Techniques

This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let $S(t)$ be the amount of salt (in grams) in the tank at time t minutes. (a) Explain why $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 0$. (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.

(a) Salt flows in at a rate of 3 grams/litre $\times 2$ litres/minute $= 6$ grams/minute. Salt flows out at a rate proportional to the current concentration $\frac{S}{200}$ grams/litre, times the outflow rate of 2 litres/minute, giving $\frac{2S}{200} = \frac{S}{100}$ grams/minute out. Net rate: $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) This is a linear first-order DE. Integrating factor: $e^{\int \frac{1}{100}dt} = e^{t/100}$. $\frac{d}{dt}(Se^{t/100}) = 6e^{t/100}$. Integrating: $Se^{t/100} = 600e^{t/100} + C$, so.

18 [Challenging] • Target: Family of solution curves and particular solutions • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Level 1 · Foundations Find the particular solution to $\frac{dy}{dx} = \frac{3x^2}{y}$ subject to $y(0) = 4$.

Separate variables: $y\,dy = 3x^2\,dx$. Integrate both sides: $\int y\,dy = \int 3x^2\,dx \Rightarrow \frac{1}{2}y^2 = x^3 + C$. Apply initial condition $y(0) = 4$: $\frac{1}{2}(4)^2 = 0 + C \Rightarrow C = 8$. Multiply by 2: $y^2 = 2x^3 + 16$. Since $y(0) = 4 > 0$, choose the positive root: $y = \sqrt{2x^3 + 16}$.

19 [Challenging] • Target: Verifying a solution to a second-order differential equation • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x + 6)] = 3$. Convert to exponential form: $x(x + 6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic: $(x + 9)(x - 3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

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20 [Challenging] • Target: Slope fields and qualitative solutions • Cross-Lesson: 6.01 Matrices (AI HL)

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

(a) Find $\frac{dy}{dx}$: $\frac{dy}{dx} = 6x^2 - 18x + 12$ Set $\frac{dy}{dx} = 0$: $6(x^2 - 3x + 2) = 0 \Rightarrow 6(x-1)(x-2) = 0$, giving $x = 1$ and $x = 2$. When $x = 1$: $y = 2(1) - 9(1) + 12(1) + 1 = 6$. Point is $(1, 6)$. When $x = 2$: $y = 2(8) - 9(4) + 12(2) + 1 = 16 - 36 + 24 + 1 = 5$. Point is $(2, 5)$. (b) Second derivative: $\frac{d^2y}{dx^2} = 12x - 18$ At $x = 1$: $\frac{d^2y}{dx^2} = 12(1) - 18 = -60$, so $(2, 5)$ is a local minimum.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving by separation of variables

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

a. $k = 5, \quad x_2 = \frac{28}{9}$ ✓

b. $k = 5, \quad x_2 = \frac{44}{9}$

c. $k = 2, \quad x_2 = \frac{14}{5}$

d. $k = 7, \quad x_2 = \frac{84}{25}$

We solve the problem in two steps: **Step 1: Find the value of k** The Newton-Raphson iteration formula is: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ Differentiating $f(x) = x^3 - 2x^2 - kx + 12$ with respect to x: $f'(x) = 3x^2 - 4x - k$ Evaluating $f(x)$ and $f'(x)$ at $x_0 = 2$: $f(2) = 2^3 - 2(2^2) - 2k + 12 = 8 - 8 - 2k + 12 = 12 - 2k$ $f'(2) = 3(2^2) - 4(2) - k = 12 - 8 - k = 4 - k$ Using the iteration formula for x_1: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow 4 = 2 - \frac{12-2k}{4-k}$ Rearran.

22 [Foundational] • Target: Modeling with differential equations

First-order linear ODE solution structure: general solution = homogeneous solution + ____

a. **particular solution** ✓

b. another homogeneous solution

c. the initial condition

d. zero

Applying Differential Equations principles, $y = y_h + y_p$. Therefore, the verified correct option is particular solution.

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23 [Foundational] • Target: First-order linear differential equations

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$ ✓

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

To solve the differential equation, we use separation of variables: 1. **Separate the variables**: $e^y\,dy = 2x\,dx$ 2. **Integrate both sides**: $\int e^y\,dy = \int 2x\,dx$ $e^y = x^2 + C$ 3. **Apply the initial condition** $y = 0$ when $x = 0$: $e^0 = 0^2 + C \Rightarrow 1 = C$ *(A frequent error is assuming $C = 0$ because the boundary values are zero, which leads to the incorrect solution $y = \ln(x^2)$.)* 4. **Express y explicitly in terms of x**.

24 [Foundational] • Target: Euler's method for numerical solutions

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$ ✓

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

The given differential equation is $\frac{dy}{dx} = 2x$. To find the general solution, we integrate both sides with respect to x: $\int \frac{dy}{dx}dx = \int 2xdx$ $y = x^2 + C$, where C is the constant of integration. We are given that the particular solution passes through the point $(1, 3)$. Substitute these values into the general solution: $3 = (1)^2 + C$ $3 = 1 + C$ $C = 3 - 1 = 2$. Therefore, the particular solution is $y = x^2 + 2$.

25 [Foundational] • Target: Second-order context: simple harmonic motion

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$ ✓

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

The integrating factor is $\mu(x) = e^{-\int \frac{1}{x}\,dx} = \frac{1}{x}$. Then $\left(\frac{y}{x}\right)' = x \Rightarrow \frac{y}{x} = \frac{x^2}{2} + C \Rightarrow y = \frac{1}{2}x^3 + Cx$.

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26 [Foundational] • Target: Family of solution curves and particular solutions

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1 e^{r_1 x} + C_2 e^{-r_1 x}$

b. $y = C_1 e^{r_1 x}$

c. $y = (C_1 + C_2) e^{2r_1 x}$

d. $y = (C_1 + C_2 x) e^{r_1 x}$ ✓

Applying Differential Equations principles, Repeated root: multiply by x. Therefore, the verified correct option is $y = (C_1 + C_2 x) e^{r_1 x}$.

27 [Foundational] • Target: Verifying a solution to a second-order differential equation

A first-order linear ODE has the form:

a. $y' + P(x)y = Q(x)$ ✓

b. $y' = P(x)y^2$

c. $y'y = Q(x)$

d. $y'' = P(x)$

Applying Differential Equations principles, Standard form: $y' + P(x)y = Q(x)$. Therefore, the verified correct option is $y' + P(x)y = Q(x)$.

28 [Foundational] • Target: Slope fields and qualitative solutions

Second-order constant-coefficient ODE $y'' - y = 0$: characteristic equation:

a. $r - 1 = 0$

b. $r^2 + 1 = 0$

c. $r^2 - 1 = 0$ ✓

d. $r^2 + r - 1 = 0$

Applying Differential Equations principles, $r^2 - 1 = 0 \Rightarrow r = \pm 1$. Therefore, the verified correct option is $r^2 - 1 = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving a linear differential equation with an initial condition

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. **maximum** ✓

d. negative

Applying Differential Equations principles, Maximum growth at $P = K/2$. Therefore, the verified correct option is maximum.

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30 [Intermediate] • Target: A logistic growth model

Linear homogeneous 2nd-order ODE: if y_1, y_2 are solutions, so is:

a. y_1^2

b. $y_1 y_2$

c. y_1'

d. $y_1 + y_2$ ✓

Applying Differential Equations principles, Superposition principle. Therefore, the verified correct option is $y_1 + y_2$.

31 [Intermediate] • Target: Checking the order and degree of a differential equation

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0 ✓

b. K

c. ∞

d. -1

Applying Differential Equations principles, At $P = K$: $dP/dt = rK(1 - K/K) = 0$. Therefore, the verified correct option is 0.

32 [Intermediate] • Target: Using Euler's method with a smaller step size

Use Euler's method with step $\frac{1}{10}$ to approximate $y(\frac{1}{5})$ for $y' = x - y$, $y(0) = 0$.

a. 0

b. $\frac{1}{25}$

c. $\frac{1}{100}$ ✓

d. $\frac{11}{100}$

Apply $y_{n+1} = y_n + h\,f(x_n, y_n)$ 2 times starting from $(0, 0)$; the last value is $\frac{1}{100}$.

33 [Intermediate] • Target: Solving by separation of variables

Method of undetermined coefficients: for $y'' + y = e^x$, try particular solution:

a. $y_p = Ae^x$ ✓

b. $y_p = Ae^{2x}$

c. $y_p = Axe^x$

d. $y_p = A\sin x$

e^x is not a solution of the homogeneous equation; try Ae^x.

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34 [Intermediate] • Target: Modeling with differential equations

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha+\beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$ ✓

Euler: $e^{\alpha x}$ envelope times sinusoids.

35 [Intermediate] • Target: First-order linear differential equations

Reduction of order: given $y_1 = e^x$ solves $y'' - y = 0$. Second independent solution via $y_2 = v(x)e^x$:

a. $y_2 = xe^x$

b. $y_2 = e^x \ln x$

c. $y_2 = xe^{-x}$

d. $y_2 = e^{-x}$ ✓

$v' + v = 0 \Rightarrow v = Ce^{-2x} \ldots y_2 = e^{-x}$.

36 [Intermediate] • Target: Euler's method for numerical solutions

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$ ✓

c. $e^{-1} - 1$

d. $2 - e^{-1}$

The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 te^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$: $y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$. $y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 te^t = (2 + c_2)e^t + (2 + c_2)te^t$.
$y'(0) = (2 + c_2)e^0 + (2 + c_2)(0)e^0 = 2 + c_2 = 1$, so $c_2 = -1$. Thus.

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37 [Intermediate] • Target: Second-order context: simple harmonic motion

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. **0.0434 ✓**

We are given the exponential growth model $P(t) = P_0 e^{kt}$. We know that the initial population $P_0 = 500$ and that after 10 years, $P(10) = 1200$. We can substitute these values into the model: $1200 = 500e^{k \cdot 10}$ Divide both sides by 500: $\frac{1200}{500} = e^{10k}$ $2.4 = e^{10k}$ To solve for k, we take the natural logarithm of both sides: $\ln(2.4) = \ln(e^{10k})$ $\ln(2.4) = 10k$ Now, divide by 10: $k = \frac{\ln(2.4)}{10}$ Using a calculator, $\ln(2.4) \approx 0.8754687$. So.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Family of solution curves and particular solutions • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347 ✓**

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0 e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

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39 [Challenging] • Target: Verifying a solution to a second-order differential equation • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$ ✓

d. $-\cos x$

Applying Differential Equations principles, $\int \tan x \, dx = -\ln|\cos x|$. Therefore, the verified correct option is $-\ln|\cos x|$.

40 [Challenging] • Target: Slope fields and qualitative solutions • Cross-Lesson: 6.01 Matrices (AI HL)

Homogeneous 2nd-order: characteristic equation $r^2 + r - 6 = 0$. Roots:

a. $r = 2, -3$ ✓

b. $r = -2, 3$

c. $r = 1, -6$

d. $r = 3, 2$

Applying Differential Equations principles, $(r + 3)(r - 2) = 0 \Rightarrow r = 2, -3$. Therefore, the verified correct option is $r = 2, -3$.

41 [Challenging] • Target: Solving a linear differential equation with an initial condition • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$ ✓

c. $y = 2x + 2$

d. $y = xe^x$

Applying Differential Equations principles, $y = Ce^x$; $y(0) = C = 2$. Therefore, the verified correct option is $y = 2e^x$.

42 [Challenging] • Target: A logistic growth model • Cross-Lesson: 6.03 Markov Chains (AI HL)

Second-order differential equation of a conic in polar: $r = \frac{1}{1 + e\cos\theta}$. At $\theta = 0$, $r =$

a. $1/(1 + e)$ ✓

b. $1/(1 - e)$

c. 1

d. e

Applying Parametric and Polar principles, $r(0) = 1/(1 + e)$. Therefore, the verified correct option is $1/(1 + e)$.

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43 [Challenging] • Target: Checking the order and degree of a differential equation • Cross-Lesson: 1.01 Proof by Mathematical Induction

The equation $y' = ky$ models exponential growth when:

a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$ ✓

Applying Differential Equations principles, $k > 0$: growth; $k0$.

44 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Resonance in $x'' + \omega_0^2 x = F_0 \cos(\omega_0 t)$: particular solution grows as:

a. $\sin(\omega_0 t)$

b. $e^{\omega_0 t}$

c. $t^2 \cos(\omega_0 t)$

d. $t\cos(\omega_0 t)$ ✓

Resonance: particular solution $\propto t\sin$ or $t\cos$.

45 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Cauchy-Euler $x^2 y'' + xy' + y = 0$: characteristic roots $r = \pm i$. Solution:

a. $y = C_1 x^i + C_2 x^{-i}$

b. $y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$ ✓

c. $y = C_1 \ln x + C_2$

d. $y = x(C_1 \cos x + C_2 \sin x)$

Complex roots: $y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$.

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